COURSE: **CHE 312 – Transport Phenomena II**

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Chicago, Illinois 60607

Overview-Objectives: Heat and mass transport phenomena. Heat conduction, convection and radiation. Heat exchanger design. Diffusion. Mass transfer coefficients.

The broad objectives of this course are:

(a) To develop clear understanding of the basic mechanisms of heat and mass transfer in physical processes encountered in daily living experiences and in chemical engineering practice; and

(b) To develop the skills to describe heat and mass processes in terms of mathematical models, make appropriate assumptions to simplify the models, and perform calculations to obtain qualitative and/or quantitative information, which can lead to better design or better understanding of the processes.

Instructor: Professor Christos G. Takoudis, SEO 207, Phone #: (312) 355-0859, Login: takoudis@uic.edu. Office Hours: W 12:00 – 1:00 PM and any time by appointment

Lecture: Class meets on TR at 2:30 – 3:45 PM, in EIB 124

Teaching Assistants: Ece Bayraktar, Login: ebayr@uic.edu Office hours: anytime except 1:45 – 3:15 PM on M & W and 9:15 – 11:45 AM on T and R, all in EIB 247.

Outline of Topics:

Introduction
Introduction to Conduction
One-Dimensional Steady-State Conduction
Two-Dimensional Steady-State Conduction
Transient Conduction

EXAM I

Introduction to Convection
Heat Transfer in External Flows

Heat Transfer in Internal Flows

Free Convection

Heat Exchangers

EXAM II

Radiation Heat Transfer

Radiation Exchange Between Surfaces

Diffusion Mass Transfer

Diffusion – Homogeneous Chemical Reaction Systems

EXAM III

Diffusion – Heterogeneous Chemical Reaction Systems

Transient Diffusion and Applications

Diffusion and Chemical Reaction Inside a Porous Catalyst

Mass Transfer Design Calculations

Multimode Heat and Mass Transfer

EXAM IV

Prerequisite: CHE 311

Homework: Approximately 12 homework assignments.

Examinations: Four one-hour exams. Conditions TBA. The approximate dates of the examinations are:

Exam I (One-Hour): After about 4 weeks of classes

Exam II (One-Hour): After about 8 weeks of classes

Exam III (One-Hour): After about 12 weeks of classes

Exam IV (One-Hour): After about 15 weeks of classes

The exact date of each exam will be announced at least one week before the exam.

There will be unannounced quizzes based on lecture material and the reading assignments. Each quiz will last 20-30 minutes. *There will be no early or make-up exams or quizzes*.

Grade: The final score for the course will be computed as follows:

*Homework Assignments**: 17 %

(***A minimum of 2/3 completed HW Assignments is required for a passing grade**. For changes to HW grades, please see your TA within 2 weeks of posting of grades. In case of illness please let your TA know as soon as possible - failure to do so will be considered as late submission and marks will be deducted.)

Unannounced Quizzes 17 %

Exam I:	17 %
Exam II:	17 %
Exam III:	17 %
Exam IV:	17 %

Textbook:

Theodore L. Bergman, Adrienne S. Lavine, *Fundamentals of HEAT and MASS Transfer*, Wiley, Eighth Edition, 2019

Other Suggested References:

Bird R., Stewart W., and Lightfoot E., *Transport Phenomena*, John Wiley & Sons, 2002 (2^{nd} Edition).